

Spectral Variation-Based Signal Compression Technique for Gapless Power Quality Waveform Recording in Smart Grids

Eder B. Kapisch , Victor V. de Moraes , Leandro R. M. Silva , *Member, IEEE*, Luciano M. A. Filho , and Carlos A. Duque , *Senior Member, IEEE*

Abstract—The current scenario of electric power systems expansion requires the insertion of new devices and technologies, contributing to the establishment of smart grids. However, the more instruments are connected to the grid, the greater the amount of data resulting from the measurements, which can cause a bottleneck in data transferring. The present article addresses this problem by proposing a system that implements a novel and efficient real-time data compression method based on the power signal spectral content variation, focused on long-term gapless oscillographic analysis. This system also provides a high-fidelity reconstruction for full oscillography power quality analysis. Comparison results with synthetic and real signals using another gapless power quality disturbance recorder present in literature and a commercial power quality disturbance recorder show that the proposed technique is capable to achieve a compression rate more than ten times greater than both while providing a better reconstruction quality, reaching compression ratio values of up to 10780:1 for synthetic signals and 9470:1 for real signals.

Index Terms—Data compression, digital signal processing, novelty detection, power quality (PQ), smart grid (SG), spectral variation.

I. INTRODUCTION

OVER the years, electric power systems (EPS) have become increasingly complex. Some of the characteristics that they commonly observed are nonlinearity and time variance, which have contributed to power quality (PQ) pollution [1].

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Eder B. Kapisch, Leandro R. M. Silva, Luciano M. A. Filho, and Carlos A. Duque are with the Department of Electrical Engineering, Federal University of Juiz de Fora, Juiz de Fora 36036-330, Brazil (e-mail: eder.kapisch@ufjf.edu.br; leandro.manso@ufjf.edu.br; luciano.andrade@ufjf.edu.br; carlos.duque@ufjf.edu.br).

Victor V. de Moraes is with the Graduate Program in Electrical Engineering - Biomedical Engineering Department, Faculty of Electrical and Computer Engineering, State University of Campinas, Campinas 13083-852, Brazil (e-mail: victor.moraes@engenharia.uff.br).

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Dealing with PQ problems, methods for identifying and minimizing these problems are indispensable, especially in the new context of smart grids (SGs), where an accurate parameter estimation, fast communication, and control of system elements are key aspects [2].

Because of the significant impact on EPS operation and its economic consequences, issues involving PQ have attracted the attention, especially from industry professionals and researchers. PQ-related works focused methods capable of detecting, identifying, and if possible, minimize consequences that a grid with a poor PQ may present have been developed [3]–[5]. Besides, in the future, unknown events may occur, and the way they interfere in the PQ is an important aspect to obtain a good performance of the system [6].

A high potential use of the measurement data lies in monitoring the condition of equipment operation, since many equipment failures present unique signatures in the voltage and current waveforms [7]. Therefore, the complete signal waveform storage composes relevant data for diagnostic, operation, protection and expansion applications, which are essential tasks in the new context of the SGs [8].

Nowadays, traditional equipment available on the market that stores EPS parameters are known as PQ analyzers. However, in general, only average values calculated within, sometimes adjustable, aggregation intervals are stored. Waveform recordings are triggered only when some disturbance is detected. The length of such continuous recordings may vary according to the disturbance duration, up to a maximum period, usually a few seconds or minutes. A recent research was conducted in this direction, gathering main PQ analyzers manufactures where one can see that most of them are not able to store gapless continuous waveform recordings [9]. Besides, one can note that the implementation of a compression stage is a determinant factor for memory and hardware optimization [10].

Although aggregation values are useful for reducing the amount of data, long period measurements, and standards compliance, a deeper analysis should be enabled by full observations of signals. While compliance with standards is required, it is not necessarily useful for EPS diagnostic purposes and detection of unknown events [11]. Thus, the loss of useful information that may be valuable for PQ maintenance should be avoided. This is also supported by the fact that the post processing of this data

can reveal information not observed previously, enabling system enhancement, troubleshooting, algorithm optimization, etc.

The most trivial way of keeping all information from data measurements is by storing all samples continuously, i.e., storing all raw data. Obviously, this requires a big storage space, which is a challenge that current digital fault recorders have been facing [12].

A more effective way to keep all relevant information from the waveform signal is by using compression methods. Several works have been published in this direction [13]. A parameter commonly used to assess the compression capacity is the compression ratio (CR), defined as the ratio between the number of bits $N_{\text{bits},x}$ required to store the original non-compressed signal $\mathbf{x}(n)$ and the number of bits $N_{\text{bits},C}$ of its compressed version [14]

$$\text{CR} = N_{\text{bits},x} / N_{\text{bits},C}. \quad (1)$$

A. Related Work

A brief review of the related works regarding PQ signals compression techniques present in the literature is given below.

Khan [15] presents a technique for denoising and lossy compression of data in SG communication using wavelet packet transform (WPT). The article uses weighted Shannon entropy to calculate the best basis of a signal from the complete WPT. A modified minimum description length algorithm is proposed to adjust the denoising threshold. The best CR value presented with satisfactory reconstruction error is 5.26:1.

In [16], a methodology for electrical power waveform signals compression using genetic algorithm and artificial neural network is proposed. CR values from 8:1 to 24:1 are achieved.

Zhang *et al.* [17] uses the sparse representation property of the wavelet transform to enhance the standard wavelet compression method. CR values of approximately 60:1 are obtained. Găspăresc [18] uses wavelet transform and spline interpolation for decomposition and reconstruction of PQ signals. A maximum CR value of 64:1 is obtained.

The work presented in [19] proposes a PQ disturbance compression method based on independent component analysis, fast Fourier transform (FFT), and an adaptive threshold obtained using mathematical morphology. CR values from 10:1 to 33.33:1 are achieved.

Some works present more than one compression stage to achieve an even higher CR. Generally, they are composed of a lossy compression stage followed by a lossless compression stage. In [20], a data compression technique for three-phase PQ signals based on wavelet transform and the Lempel–Ziv–Welch (LZW) coding stage is presented. CR values higher than 300:1 are achieved. Only simulation results are provided.

Ruiz *et al.* [21] propose a high-performance compression algorithm for PQ signals using wavelet transformation (WT) with amplitude, ripple, and shifting corrections. The best CR is 507.61:1.

The works cited so far have two characteristics in common. First, they are applied to previously detected disturbance signals segments, not long-term gapless oscillography signals. Thus,

the application of these techniques becomes possible once the disturbances have already been detected. Second, none of them show concern at unknown events, which may occur in the new scenario of SGs.

The work proposed in [22] is focused on gapless recording and proposes the concept of novelty detection, which identifies any event, even if it is unknown. Three compression stages are used. The first one is the novelty detection concept acting as a compressor because only novelty frames are selected to represent the whole signal. The second stage applies a WT, and the final stage uses the LZW coding. The best CR value achieved in real signal tests is 570:1. These are the reasons why [22] has been taken as a reference goal to be achieved and surpassed. In the literature review, the mentioned work has the best results so far.

However, Silva *et al.* [22] applies the novelty detection in the time domain, so that many samples from novelty segments need to be saved to allow the signal reconstruction. This weakness can be overcome by a frequency-domain detection, where only the bins related to new components must be saved, improving CR.

It is worth mentioning that the reconstructed signal after decompression must preserve its main characteristics to provide subsequent analyzes and diagnoses. In this context, this paper proposes a spectral variation detection and compression system (SVDCS), which is able to perform a gapless waveform recording with high fidelity reconstruction. The SVDCS proposed in the present paper can be useful in long-duration gapless data storage and oscillographic analysis applications, commonly present in data loggers, power analyzers, and PMUs equipment [23].

B. Main Contributions

The proposed system implements two stages of compression, one lossy and one lossless, achieving high compression rates. Besides, the proposed system is compared to another waveform gapless power quality disturbance recorder (G-PQDR) [22], and to a commercial power quality disturbance recorder (C-PQDR) as well [24].

Thus, the main contributions of this article are as follows:

- 1) A novel signal compression technique based on the spectral variation, with the description of the compression and the decompression schemes.
- 2) Contribute to long-term PQ data storage applications.
- 3) Performance comparison of the proposed technique with other existing compression schemes from the literature [22] and a commercial equipment [24].

C. Paper Organization

The rest of this article is organized as follows. Section II presents the proposed compression system. Section III describes the reconstruction process for the proposed system. Section IV provides the hardware description. Sections V and VI bring some results and discussion, respectively. Finally, Section VII concludes this article.

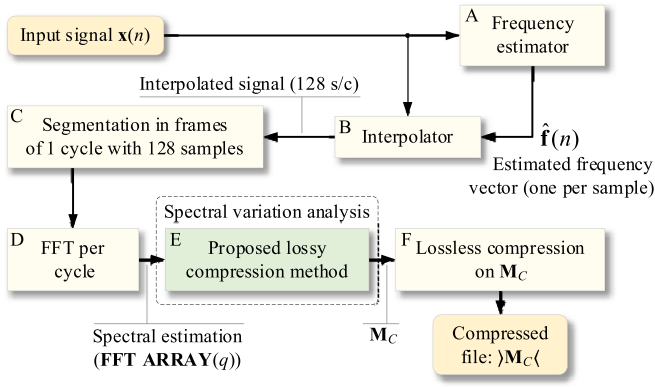


Fig. 1. Proposed compression system (SVDCS) based on spectral variation analysis.

II. PROPOSED COMPRESSION SYSTEM

In the proposed compression system, the amount of lost data during the lossy compression stage is controlled by a thresholding process applied in the frequency domain. However, relevant information for PQ analysis is always kept. A complete lossless compression operation is achieved by simply setting all thresholds to zero.

The spectral content is estimated through the FFT of each fundamental component cycle, which corresponds to the spectral analysis frame. The novelties are detected in the frequency domain, by comparing FFT bins. The choice of FFT is because it is one of the most common tools used in spectral estimation applications present in the literature [25].

Fig. 1 shows the flow diagram of the proposed compression system (SVDCS). The input of the system is a high-sampled single-phase discrete time signal represented as a vector indexed by position n , named $\mathbf{x}(n)$. Each sample value of the vector $\mathbf{x}(n)$ is represented by the notation x_n , $n = 1, 2, \dots, N$. Thus, $\mathbf{x}(n) = [x_1, x_2, \dots, x_n, \dots, x_N]$ is a row vector.

The compression system includes some signal preparation steps before the spectral variation analysis, such as frequency estimation, interpolation and FFT. Each part is explained below.

A. Frequency Estimator

The frequency estimator provides the fundamental frequency estimation $\hat{f}(n)$ of the input signal $\mathbf{x}(n)$, required for signal interpolation. For each sample x_n that comes from the input signal vector $\mathbf{x}(n)$, there is one frequency estimation value $\hat{f}_n$. The fundamental frequency estimator used in this article is based on zero-crossing (ZC) [26]. The choice of using a ZC-based technique is due to its simplicity and efficiency, which contributes to facilitate the hardware implementation.

B. Interpolator

A synchronous sampling must be performed on the input signal $\mathbf{x}(n)$ so that the spectral estimation provided by the FFT does not contain estimation errors, such as spectral leakage or the picket fence effect. The interpolator is used for this

purpose, which ensures that each frame with one fundamental cycle duration has exactly N_{ppc} samples, where N_{ppc} is a power of two for the FFT calculation. Thus, the sampling frequency of the input signal is not a restriction for the method, which accepts any value, even a noninteger one. This is possible due to the presence of the interpolator. High sampling rates can be used to ensure that high-order spectral components, such as high-frequency harmonics, interharmonics and transients are captured, preserving all useful information for PQ monitoring [27].

The interpolator used in this article is based on Lagrange polynomial interpolation. This choice is due to the ease of a real-time implementation as a filter bank [28].

C. Segmentation

Since the interpolator ensures that there are N_{ppc} samples in each cycle, a simple sample counter is sufficient to segment the interpolated signal into N_{ppc} sample fixed-length frames. Then, the content is loaded into the next part, which is the FFT calculation, followed by the rest of the compression algorithm.

D. FFT Per Cycle

Each frame in the SVDCS has one fundamental cycle length (N_{ppc} samples). Thus, the FFT per cycle results in a vector with N_{ppc} complex bins, each one representing a harmonic component according to the FFT resolution. In this case, the resolution is equal to 60 Hz, once there is one integer cycle in the FFT window.

The SVDCS deals with the real and imaginary parts of each complex bin separately. To do that, it stores these parts in a row vector named **FFT_ARRAY**(q), where each q -position refers to a real or imaginary part. So, the length of **FFT_ARRAY**(q) is $2 \times N_{ppc}$ q -positions, as expressed in (2). In the adopted notation, $\top$ is the transposition operator. Each value of **FFT_ARRAY**(q) is represented by fft_array_q , where $q = 1, 2, \dots, 2 \times N_{ppc}$.

$$\mathbf{FFT_ARRAY}(q)_{q=1, \dots, Q=2 \times N_{ppc}}^\top = \begin{bmatrix} \text{Re}(1^{\text{st}} \text{bin}) \\ \text{Im}(1^{\text{st}} \text{bin}) \\ \vdots \\ \text{Re}(N_{ppc}^{\text{th}} \text{bin}) \\ \text{Im}(N_{ppc}^{\text{th}} \text{bin}) \end{bmatrix}^\top = \begin{bmatrix} \text{fft_array}_1 \\ \text{fft_array}_2 \\ \vdots \\ \text{fft_array}_{2 \times N_{ppc} - 1} \\ \text{fft_array}_{2 \times N_{ppc}} \end{bmatrix}^\top. \quad (2)$$

E. Spectral Variation Lossy Compression Method

The spectral variation analysis block (block E in Fig. 1) is the core of the system, because it is responsible to eliminate redundancies present in the signal spectra. It analyzes and compares the **FFT_ARRAY**(q) vector of each frame trying to find any variation, which indicates the presence of electrical disturbances. If there is no variation, the system calculates the mean value of each q -position and the number of frames where they repeat, resulting in signal compression. The spectral variation compression method is explained in details below. Fig. 2 shows its structure.

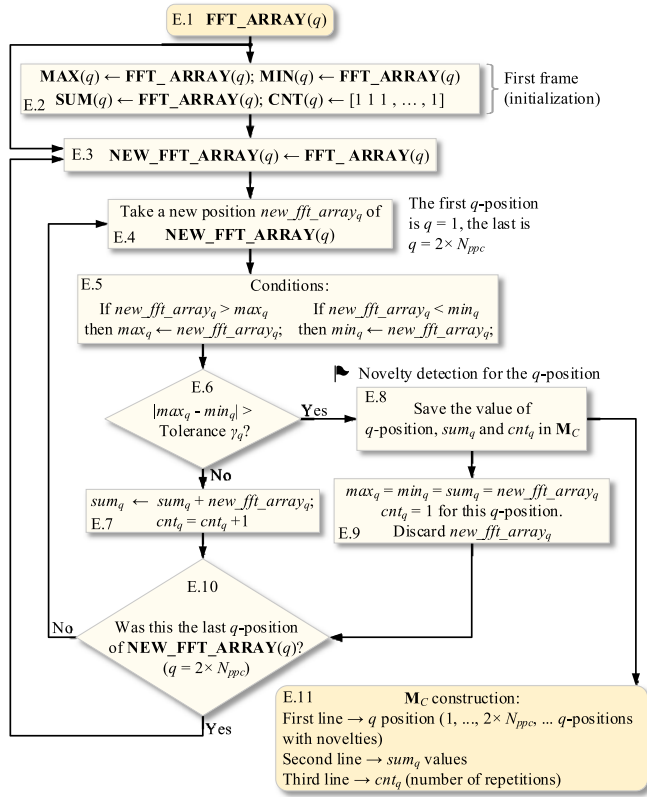


Fig. 2. Proposed spectral variation compression method, part of the proposed SVDSC (block E of Fig. 1).

The proposed compression method takes the $2 \times N_{ppc}$ - position **FFT_ARRAY**(q) vector of each frame as the input (block E.1 in Fig. 2).

In order to make the frame-by-frame comparison, the algorithm uses other vectors to store the maximum, minimum and accumulated value after each frame comparison. These vectors are named **MAX**(q), **MIN**(q), and **SUM**(q), respectively, both with the same size as **FFT_ARRAY**(q), that is, $2 \times N_{ppc}$ q -position length, related to real and imaginary parts of each complex bin. To store the number of times that each real and imaginary part repeat along the time, there is another vector named **CNT**(q).

Each element of the vectors **MAX**(q), **MIN**(q), **SUM**(q), and **CNT**(q) follows the same notation as the one used in (2) for **FFT_ARRAY**(q), that is, $\max_q$, $\min_q$, sum_q , and cnt_q .

The first step is the frame initialization, when both **MAX**(q), **MIN**(q), and **SUM**(q) receive the values from **FFT_ARRAY**(q). This is done because at the beginning there is no previous frame to make a comparison. The **CNT**(q) is initialized with all q -positions equal to 1, since it is the first frame. This step is represented by Block E.2 in Fig. 2.

After the frame initialization, the system waits for a new **FFT_ARRAY**(q), and save it in a vector named **NEW_FFT_ARRAY**(q) (E.3 in Fig. 2), which has each element represented by new_fft_array_q (see E.4 in Fig. 2).

For each q -position, the element new_fft_array_q is compared to the elements $\max_q$ and $\min_q$ to verify if the new frame has

any value higher than the correspondent position previously-stored in **MAX**(q) or any value lower than the correspondent position previously-stored in **MIN**(q). If the value of any q -position is identified as higher than the respective position of **MAX**(q) or lower than the respective position of **MIN**(q), this position is updated in **MAX**(q) or **MIN**(q) vectors. This process is represented by the block E.5 in Fig. 2.

With the **MIN**(q) and **MAX**(q) vectors updated, a threshold is applied to the comparison of the absolute value of the difference between them. There is one threshold γ_q for each q -position with a tolerance vector $\Gamma(q)$ is performed (see E.9 in Fig. 2). There is a different threshold value γ_q for each q -position of $\Gamma(q)$, according to the notation adopted. The explanation for different threshold values is described at the end of this section.

If the absolute value of the difference between $\max_q$ and $\min_q$, is smaller than the tolerance threshold γ_q for a given q -position (“no” option from E.6, Fig. 2), it means that there is no novelty and the vectors **SUM**(q) and **CNT**(q) are updated on this q -position in the following way: the value of new_fft_array_q is summed to sum_q , and cnt_q is incremented by 1 (E.7 in Fig. 2). Therefore, it is possible to calculate the mean value of each q -position by dividing each element of **SUM**(q) by each correspondent element of **CNT**(q). However, this operation can be performed during the decompression process (offline) without interfering on the functionality of the proposed method.

If the absolute value of the difference between $\max_q$ and $\min_q$ is greater than γ_q for the considered q -position (“yes” option from E.6, Fig. 2), a novelty detection for this q -position is flagged. This indicates that there is a variation in that q -position of the **NEW_FFT_ARRAY**(q), and all the information for the signal reconstruction must be saved, which are the q -position index, **SUM**(q) value for this q -position, and **CNT**(q) value for the same position (E.8 in Fig. 2). This novelty detection step ensures that all relevant information about any electrical disturbance present in the analyzed signal is kept for later reconstruction and analysis, regardless of the disturbance intensity or duration.

After saving this information, $\max_q$, $\min_q$, and sum_q are updated for this q -position in its respective vectors with the same value for new_fft_array_q (E.9 in Fig. 2). If the current q -position is not the last position of **NEW_FFT_ARRAY**(q) (“no” option from E.10), the next position is analyzed from E.4 to E.9 until the whole **NEW_FFT_ARRAY**(q) vector is finished (E.10 in Fig. 2). After the analysis of **NEW_FFT_ARRAY**(q) is completed, the next **NEW_FFT_ARRAY**(q) vector corresponding to the next new cycle is taken and the comparison process restarts (from E.3 in Fig. 2).

The information saved in the step shown by E.8 is organized as a three-row compression matrix, named **M_C**, according to the following row convention. The first row of **M_C** is composed of all the q -position indexes where any novelty is detected in the frequency domain. The second row is composed of the sum_q value for each q -position of the first row. Finally, the third row is composed of cnt_q values (E.11 in Fig. 2). That way, the number of columns of **M_C** is not fixed, but depends on the number of events present in the signal. For each novelty detected for a given q -position, a new column is concatenated to the right side of **M_C** following the given row convention. The minimum number of

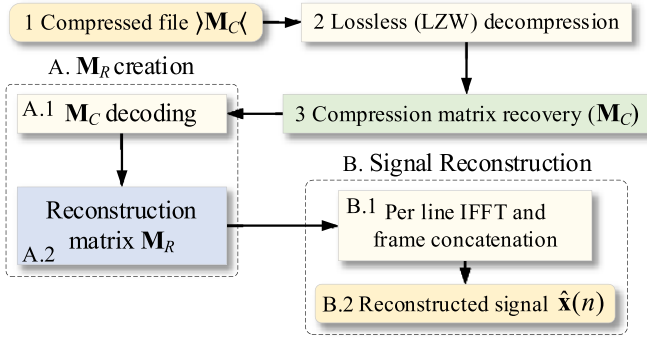


Fig. 3. SVDCS Reconstruction process.

columns is N_{ppc} , which represents one cycle being repeated the same along the time.

Generally, in power signals, the energy of the harmonic components is less than the energy of the fundamental component. Therefore, it is more sensible to use more sensitive values for thresholds applied to the bins of higher harmonic orders instead of a single value for all bins. The higher the order, the lower should be the threshold value applied, that is, the more sensitive to the variation of this spectral component is the system. Such a strategy increases the accuracy of the method and improves the quality of signal reconstruction, ensuring that even the highest order components, whether they are of a lesser or greater degree of intensity, are captured. In order to accomplish this strategy, the variation of the threshold values γ_q follows a decreasing exponential distribution as a function of the frequency bin. Thus, the value for each threshold γ_q to form the tolerance vector $\Gamma(q)$ is given by the generating function expressed in

$$\gamma_q = Gq^{-\beta} \quad (3)$$

where G is a gain factor, q is the q -position index of $\Gamma(q)$, and $\beta < 1$ is the exponent responsible for the exponential variation of the tolerance values.

F. Lossless Compression

The last step of the proposed SVDCS consists of a lossless compression applied to the compression matrix $\mathbf{M}_C$. After this step the final compressed file symbolized by $\mathbf{M}_C$ is created, optimizing the memory space.

The well-known LZW algorithm was chosen for the lossless compression stage [29], as it offers high compression rates and it has open-source distributions, such as 7-zip [30], since December 2008. After compression, the $\mathbf{M}_C$ file can be stored in a storage device for later decompression.

III. RECONSTRUCTION PROCESS

The SVDCS signal reconstruction process is shown in the flow diagram depicted in Fig. 3. It is basically the inverse process to the compression process described in Section II, and it can be done offline. Each step is explained below.

The reconstruction process starts with the $\mathbf{M}_C$ compressed file (1 in Fig. 3), resultant from the last compression step. The

first step is to perform a lossless decompression with the LZW algorithm (2 in Fig. 3). After this step, the compression matrix $\mathbf{M}_C$ is recovered (3 in Fig. 3).

The next steps are the reconstruction matrix $\mathbf{M}_R$ creation (A.1 and A.2 in Fig. 3) and the signal reconstruction (B.1 and B.2 in Fig. 3). These steps are explained as follows.

A. Reconstruction Matrix

This part of reconstruction process consists of decoding the data that has been stored in the compression matrix $\mathbf{M}_C$ in order to create a reconstruction matrix, named $\mathbf{M}_R$ (A.1 and A.2 in Fig. 3). Each row of $\mathbf{M}_R$ has $2 \times N_{ppc}$ q -positions, corresponding to the real and imaginary parts of each of the N_{ppc} complex bins of the Fourier spectrum of each one-cycle frame of the signal $\mathbf{x}(n)$. Considering that $\mathbf{x}(n)$ has N_{frames} one-cycle frames, the matrix $\mathbf{M}_R$ will have N_{frames} rows. Thus, $\mathbf{M}_R$ has a dimension of $[N_{frames} \times 2 \times N_{ppc}]$.

B. Signal Reconstruction

To recover the complex bins of each one-cycle spectrum from $\mathbf{M}_R$ rows, just add the real and imaginary parts ($\text{Re} + j\text{Im}$) of each bin, forming complex bins. Once we have the bins of the FFT per cycle (from each row of $\mathbf{M}_R$), just perform the inverse transformation (IFFT) to recover each cycle in the time domain (B.1 in Fig. 3).

After recovering all recovered one-cycles frames, it is necessary to concatenate the result, cycle by cycle, to form the reconstructed signal, named $\hat{\mathbf{x}}(n)$ along the time (B.2 in Fig. 3). At this point, $\hat{\mathbf{x}}(n)$ is ready to be plotted, sample by sample, finishing the reconstruction process.

C. Reconstruction Quality Metrics

Considering the original and the reconstructed signals represented by the vectors $\mathbf{x}(n) = [x_1, x_2, \dots, x_n, \dots, x_N]$ and $\hat{\mathbf{x}}(n) = [\hat{x}_1, \hat{x}_2, \dots, \hat{x}_n, \dots, \hat{x}_N]$, respectively, both with length N , where $n = 1, \dots, N$ symbolizes the sample index of the each discrete-time signal according to the adopted notation, there are some metrics to evaluate the quality of reconstruction. The following quantities are used, the normalized mean squared error (NMSE) defined in (4), the cross correlation (COR) between the reconstructed signal and the original signal defined in (5), where the operator “.” is the inner product between two vectors, and the percentage of retained energy [RTE(%)], defined in (6) [31]

$$\text{NMSE} = \frac{\sum_{n=1}^N |x_n - \hat{x}_n|^2}{\sum_{n=1}^N |x_n|^2} \quad (4)$$

$$\text{COR} = \frac{(\mathbf{x}(n)^\top \cdot \hat{\mathbf{x}}(n))}{(\mathbf{x}(n)^\top \cdot \mathbf{x}(n))} \quad (5)$$

$$\text{RTE}(\%) = 100\% \times \frac{\sum_{n=1}^N \hat{x}_n^2}{\sum_{n=1}^N x_n^2} \quad (6)$$

The closer to 0 is the NMSE, the closer to 1 is the COR, and the closer to 100% the RTE(%) the better the signal reconstruction.

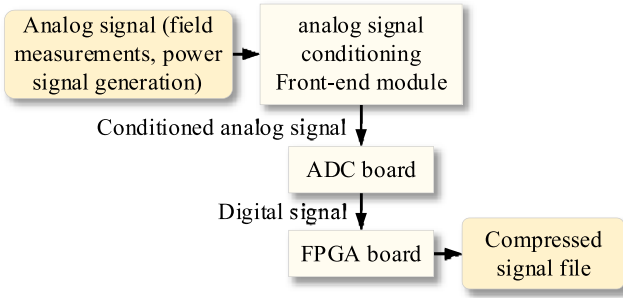


Fig. 4. SVDCS prototype diagram.

IV. HARDWARE DESCRIPTION

This section describes the SVDCS prototype for real-time implementation and the setup for experimental tests.

A. SVDCS Prototype

A hardware prototype was built to implement the SVDCS. The prototype consists of three main modules. The front-end analog module, the analog to digital converter (ADC) module, and the FPGA development kit module, as shown in Fig. 4. The front-end conditioning module board is responsible for voltage and current signal conditioning from field levels to processing levels. The ADC board converts the signals and transmit the information to the FPGA board, where all hardware implementation was made.

We built both the front-end analog and ADC boards. The former is able to condition four voltage and four current channels. The latter has a AD7606 ADC from ANALOG DEVICES, which has up to 8 simultaneous channels with 16-b resolution and supports a sampling frequency up to 200 kHz.

The FPGA development kit used to make the hardware implementation was the terasIC/ALTERA Cyclone V GX Starter Kit, which has a Cyclone V GX 5CGXFC5C6F27C7N FPGA device with 77 K programmable logic elements, 4884 Kbits of embedded memory, six fractional PLLs, two hard memory controllers, among other features. The kit is capable of running at a frequency of up to 200 MHz.

An important point to highlight is that it was possible to implement the proposed SVDCS using few resources of a low-cost FPGA-based platform, which shows the feasibility of the implementation. For this implementation, only 13 860 logic elements (18% of total) running at 25 MHz were needed to implement all SVDCS algorithms.

B. Setup

To generate the PQ disturbance power signals, we used an OMICRON CMC 256-6 plus power generator, which can be used to test and calibrate PQ measurement devices of class A and S. Some of its features include 4×300 V or 2×600 V voltage outputs and 6×12.5 A or 3×25 A or 1×75 A current outputs.

To visualize and monitor the generated waveforms we used the keysight MSOX3012T dual channel oscilloscope with 100 MHz bandwidth and maximum sampling rate of 5 GSa/s.

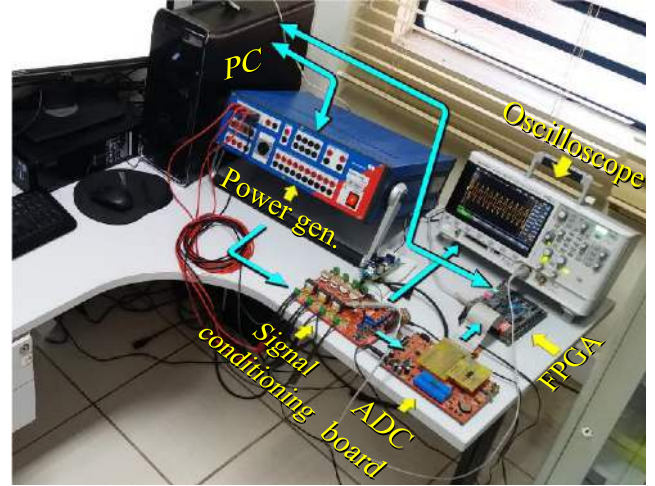


Fig. 5. SVDCS real-time implementation setup.

To control the setup equipment and run simulation tests, we used a personal computer.

A picture of the setup is shown in Fig. 5. The labels indicate the setup equipment. The cyan arrows show the communication and signal flux.

V. RESULTS

This section presents compression/reconstruction-related simulation and experimental tests to evaluate the proposed SVDCS.

First, we conducted long-duration synthetic signal tests to compare SVDCS and G-PQDR performances. Second, sampling rate variation tests were applied to SVDCS to evaluate its behavior for four different values of N_{ppc} . Third, a compression and reconstruction comparison using real signals performed. Finally, prototype tests were done through hardware simulation and using a long-term real signal measurement in order to show the compression performance comparison between SVDCS, G-PQDR, and C-PQDR.

In order to perform the synthetic signal tests, we used MATLAB software. The prototype experimental tests were performed with the setup described in previous section.

A. Long-duration Synthetic Signals

These tested signals are the same as those used in [22]. The tolerance parameters from) used during the tests were $G = 0.03$ and $\beta = 0.1$. These values were chosen empirically considering the tradeoff between the CR and the quality of reconstruction.

The signal samples were quantized using a 16-b resolution in order to emulate the 16-b AD converter of the setup ADC board. The real and imaginary parts of the FFT complex bins were also quantized with 16 b. An $N_{ppc} = 128$ samples per cycle (s/c) was adopted in this part in order to correspond the same value used for G-PQDR. Considering a nominal fundamental frequency equal to 60 Hz, the sampling frequency is $F_s = 7.680$ kHz, and the sampling period is $T_s \approx 130 \mu s$.

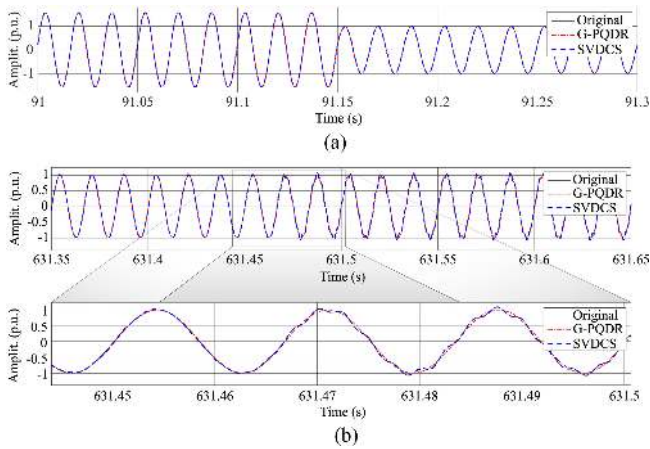


Fig. 6. Case 1 reconstruction. (a) End of a swell disturbance with intensity of 1.5 p.u. (b) Start of time-varying interharmonics. In both (a) and (b), the original signal (black solid line), the signal reconstructed by G-PQDR (red dot-dashed line), and the signal reconstructed by SVDSCS (blue dashed line).

As the three reconstruction quality metrics described in Section III are normalized by the original signal samples in denominator, and are independent of the absolute value of the amplitude (Volts, Amps), the graphical representation of the reconstructed synthetic signals is normalized by the amplitude of the fundamental component (p.u.).

Several disturbances were artificially created, such as sags, swells, interruptions, transients, time-varying frequency, etc. However, for the sake of space, only the four most representative cases are shown in this part. Each case has more than one type of PQ disturbance.

- 1) *Case 1:* In this case, the signal is composed of different disturbances interleaved with a pure nominal sinusoidal signal. The disturbances present in the signal are sags, swells, even and odd harmonics, interharmonics, notches, and ramp magnitude variations. The signal duration is about 13 min.

To store the entire signal continuously with the adopted sampling rate of 7680 Hz with 16-b resolution per sample, a 12.15 MB file would be necessary. The size of the file compressed by the G-PQDR is 102.29 kB, constituting a CR of approximately 119:1. The size of the compressed file generated by the C-PQDR for the same test signal is 254.34 kB, hence the CR is 46:1. The file compressed by the SVDSCS requires only 1.4 kB to be stored, which is a CR of about 8460:1.

The reconstruction quality metrics for G-PQDR are $NMSE = 0.0343$, $COR = 0.991$, and $RTE(\%) = 98.37\%$, while for SVDSCS are $NMSE = 1.7341 \times 10^{-4}$, $COR = 0.99969$, and $RTE(\%) = 99.97\%$.

Fig. 6(a) and (b) depict two moments of Case 1 signal. Fig. 6(a) shows the moment where a swell disturbance ends. We can observe that both compression systems are capable of tracking the dynamics of the original signal during the swell event. Fig. 6(b) shows the starting moment of a time-varying interharmonic content. Interharmonic disturbances are,

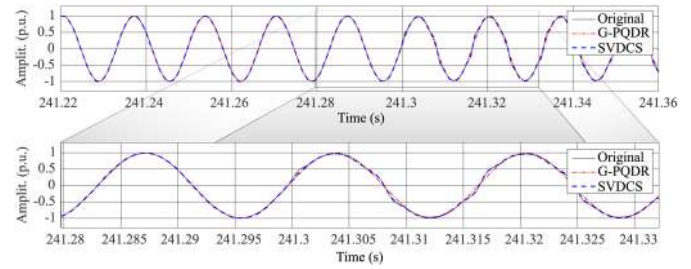


Fig. 7. Case 2 reconstruction. The beginning of a time-varying harmonic distortion.

in general, quite challenging to deal. We can see in the zoom at the bottom of Fig. 6 that the SVDSCS was able to reconstruct the signal during the interharmonics event more accurately than the reconstruction provided by G-PQDR.

It is possible to note that the SVDSCS is able to reconstruct the case 1 signal with higher quality than the G-PQDR, achieving a CR about 70 times higher than G-PQDR and 180 times higher than C-PQDR.

- 2) *Case 2:* This signal is composed of 2 min of harmonic distortion (by even harmonics or odd harmonics) interleaved with a pure sinusoidal signal for 1 min. The signal duration is about 10 min, and the size needed to store the signal without compression would be 9.25 MB. The signal compressed by the G-PQDR resulted in a 16.22 kB file, constituting a CR = 570:1. The size of the file compressed by the C-PQDR is 36.00 kB, with a CR = 250:1. The SVDSCS compressed file requires only 857 bytes, with a CR = 10780:1.

The reconstruction quality metrics for G-PQDR are $NMSE = 0.0270$, $COR = 0.992$, and $RTE(\%) = 98.54\%$, while for SVDSCS are $NMSE = 1.9633 \times 10^{-4}$, $COR = 0.99971$, and $RTE(\%) = 99.97\%$.

For this case, SVDSCS is able of a signal reconstruction with higher quality than the G-PQDR, achieving a CR about 19 times higher than G-PQDR and 43 times higher than C-PQDR.

Fig. 7 shows a part of Case 2 signal containing a time-varying harmonic distortion. The bottom of this figure shows the event beginning transition zoomed. In this case, we can see that the SVDSCS was able to reconstruct the signal more accurately.

- 3) *Case 3:* This signal is composed of a pure sinusoidal signal with a ramp magnitude variation. The magnitude varies between 127 and 180 V (nominal amplitude). To store the entire signal (10 min duration) without compression it would require a 9.1 MB file. The file compressed by the G-PQDR has 20.22 kB, achieving a CR of 448:1. The size of the file compressed by the C-PQDR is 22.67 kB, with a CR = 435:1. The size of the file compressed by SVDSCS is only 769 bytes, which corresponds to a CR = 11700:1.

The reconstruction quality metrics for G-PQDR are $NMSE = 0.0277$, $COR = 0.989$, and $RTE(\%) = 97.8\%$, while for SVDSCS are $NMSE = 1.385 \times 10^{-4}$, $COR = 0.99982$, and $RTE(\%) = 99.98\%$.

Since amplitude variations are very slow, there is no reason to plot these curves. However, we can see from the reconstruction

TABLE I
COMPRESSION RESULTS FOR SYNTHETIC SIGNALS

Signal	Case 1	Case 2	Case 3	Case 4
Original size	12.15 MB	9.25 MB	9.06 MB	1.32 MB
G-PQDR	102.29 kB	16.22 kB	20.22 kB	28.34 kB
C-PQDR	254.34 kB	36.00 kB	22.67 kB	122.2 kB
SVDCS	1.4 kB	857 B	769 B	728 B
CR G-PQDR	119:1	570:1	448:1	47:1
CR C-PQDR	46:1	250:1	435:1	11:1
CR SVDCS	8460:1	10780:1	11700:1	1807:1

TABLE II
RECONSTRUCTION QUALITY RESULTS FOR SYNTHETIC SIGNALS

Signal	Case 1	Case 2	Case 3	Case 4
NMSE G-PQDR	0.343	0.0270	0.0277	0.07
NMSE SVDCS	1.74×10^{-4}	1.96×10^{-4}	1.39×10^{-4}	4.48×10^{-4}
COR G-PQDR	0.991	0.992	0.989	0.992
COR SVDCS	0.99969	0.99971	0.99982	0.9999993
RTE G-PQDR	98.37%	98.54%	97.8%	98.85%
RTE SVDCS	99.97%	99.97%	99.98%	99.99993%

metrics that the SVDCS is able to reconstruct the case 3 signal with higher quality than the G-PQDR, achieving a CR about 26 times higher than G-PQDR and C-PQDR.

- 4) *Case 4*: This signal is a pure sinusoidal signal with a fundamental frequency variation. The fundamental frequency varies between 59 and 61 Hz. The duration of the stored signal is 86 s and the file size without compression would be 1.32 MB. The G-PQDR compressed file has 28.34 kB, constituting a CR of 47:1. The size of the compressed file by the C-PQDR is 122.2 kB, which represents a CR = 11:1. The file compressed by SVDCS requires only 728 bytes to be stored, with a CR = 1807:1.

The reconstruction quality metrics for G-PQDR are NMSE = 0.07, COR = 0.992, and RTE(%) = 98.85%, while for SVDCS are NMSE 4.48×10^{-4} , COR = 0.99999933, and RTE(%) = 99.999933%.

We can observe that the SVDCS is able to reconstruct the case 4 signal with higher quality than the G-PQDR, achieving a CR about 48 times higher than G-PQDR and 164 times higher than C-PQDR.

Tables I and II give summaries of the synthetic signals compression and reconstruction results, respectively. We can observe that in all tested cases SVDCS outperforms G-PQDR and C-PQDR.

B. Sampling Rate Variation Tests

In order to evaluate the performance of the SVDCS for different values of sampling rates F_s , a series of tests was carried out with synthetic signals containing the most common PQ disturbances: transients; sags; swells; harmonics; interharmonics; interruptions; notches; spikes; phase shifts; dc levels; and fundamental frequency variation.

A total of 100 signals were generated per disturbance. The length of each signal was 70 fundamental cycles and at least one event was present in each signal. All disturbances were generated randomly in both intensity and location. A signal-to-noise ratio

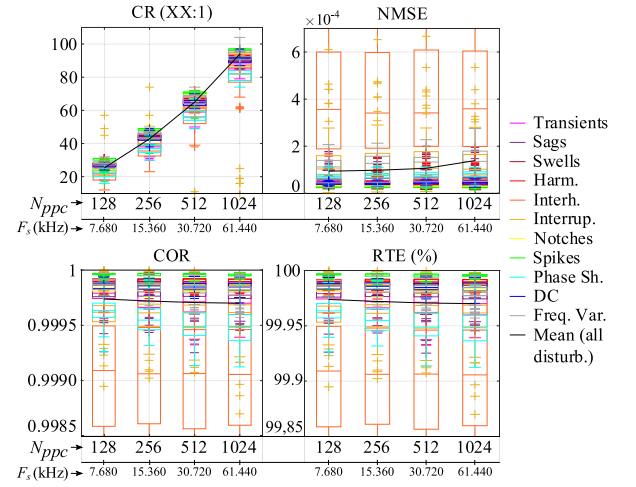


Fig. 8. SVDCS sampling rate variation tests. Up-left: compression rates. Up-right: NMSE. Down-left: COR. Down-right: RTE.

(SNR) of 45 dB was also used to approximate the simulation to real power signal measurements.

The variation in F_s was achieved by changing the value of N_{ppc} . Four different values of N_{ppc} were used for each signal: $N_{ppc} = 128$, which corresponds to $F_s = 7.680$ kHz, $N_{ppc} = 256$ ($F_s = 15.360$ kHz), $N_{ppc} = 512$ ($F_s = 30.720$ kHz), and $N_{ppc} = 1024$ ($F_s = 61.440$ kHz). It is important to remark that at this point any value of F_s could be possible, including non-integer values. These values were chosen for the sake of comparison with other approaches.

The same long-duration synthetic signals tests tolerance parameters were adopted.

For all 100 signals of the same disturbance, CR, NMSE, COR, and RTE were calculated for each value of N_{ppc} . Fig. 8 shows the variation of these four parameters along the variation of N_{ppc} (or F_s).

We can see that SVDCS presents compression rates and reconstruction quality metrics very similar, regardless of the disturbance kind. Besides, the higher the sampling rate for the same signal, the higher its compression rate. This is because, in the frequency domain, the same signal has only the same bins, regardless of the sampling rate, while in the time domain, a high sampling rate inevitably leads to bigger files for sample-by-sample non-compressed storage.

Regarding the reconstruction quality metrics, all presented good performances independent on F_s . The disturbances that showed the worst performance were the interharmonics. This can be explained by the fact that, as the window used to calculate the FFT was only 1 cycle, the frequency resolution was 60 Hz. When the interharmonic signals were generated, any value of frequency components were allowed, even non-integer ones. So, the method uses all other harmonic bins available to represent the signal.

It is important to observe that in both long-duration synthetic signal tests and sampling rate variation simulation tests, we used signals with disturbances representing real phenomena considering actual PQ indexes.

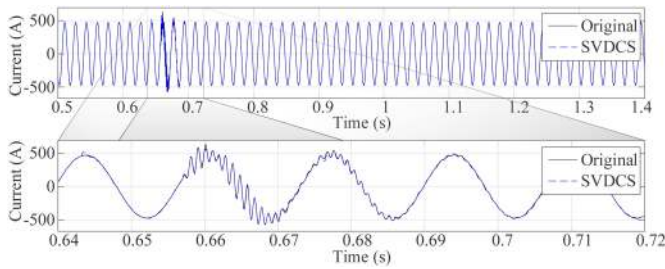


Fig. 9. Real signal compression.

C. Real Signal Compression Comparison

In order to compare the reconstruction of synthetic signals with real signals, we used real power signal measurements from a 230 kV nominal voltage substation. The sampling rate is $F_s = 15360$ kHz ($N_{ppc} = 256$). Fig. 9 shows a transient disturbance present on the current waveform of phase A. The length of the signal is 76 cycles, which is similar to the length of the synthetic signals used in the previous subsection. For this case, the results are $CR = 43$, $NMSE = 4.94 \times 10^{-5}$, $COR = 0.99989$, and $RTE = 99.9889\%$. These values are very close to the ones obtained from simulations.

A. Prototype Tests

In order to evaluate the functionality of the prototype, the following tests were performed.

1) **Hardware Simulation:** Fig. 10 shows a register transfer level simulation of the FPGA performing the compression process. The simulation was performed in ModelSim software. The value of N_{ppc} used in this simulation was $N_{ppc} = 128$, which corresponds to 256 q -positions.

We can see the waveform of the analyzed signal on the first line from top to bottom. This signal is composed of the fundamental component $f_1 = 60$ Hz and a time-varying harmonic. After four cycles with only the fundamental frequency component, the seventh harmonic with an amplitude of 0.2 p.u., is added for another three cycles until the signal ends. The signal visualization output bus has 16 b. The signal was quantized in 16 b, leaving three unoccupied bits to avoid overflow. Hence, the maximum amplitude of the fundamental the component is 4096, as shown in Fig. 10.

In the bottom part of the figure, we can see the creation of the M_C compression matrix by the last three lines out 1, out 2, and out 3, which correspond to the q -position, SUM, and CNT lines of M_C , respectively. We can notice that q values vary from 0 to 255, because of the FPGA indexing, which starts in 0 instead of 1.

2) **Long-Duration Real Signal:** The signal used in this case was acquired from a wind generator. There are eight acquisition channels, four voltages and four currents. The signal duration is 34 h. Fig. 11(a) shows a one-hour length window of one of the four measured current channels [32]. This part of the signal was applied to the setup described in Section IV.

The graphical representation of the one-hour length window in Fig. 11(a) does not allow us to visualize the necessary details to evaluate the waveform. However, this figure is intended to reveal that, sometimes, current signals might present chaotic behavior when compared to voltage signals. Even though, the reconstructed signal must be capable of following the amplitude oscillations.

An additional reconstruction metric that can be used to show the reconstruction error along the time for very long signals such as in this case is the normalized absolute error (NAE). This metric shows whether or not the reconstruction maintains consistency in quality along the time. The normalization is relative to the nominal root mean square (RMS) voltage. The NAE is defined in (7). A good reconstruction quality keeps the NAE close to 0% along time

$$NAE = 100\% \times |x_n - \hat{x}_n| / V_{RMS}. \quad (7)$$

We can see in the bottom plot of Fig. 11(a) that the SVDSCS NAE remains close to zero along the time (below 0.02), while G-PQDR NAE are greater than SVDSCS NAE along the time, and presents oscillation. This oscillation in the error presented by the G-PQDR is due to the gradual phase deviations that its reconstruction presents during some moments. SVDSCS is immune to this type of error due to its methodology, in which no phase shift exists. This is another contribution of this work. Fig. 11(b) shows a zoom with a harmonic distortion.

Table III gives the results of compression and reconstruction quality obtained by SVDSCS, G-PQDR, and C-PQDR for the voltage and current channels, separately.

We can observe from Table III that SVDSCS was able to achieve a maximum CR more than ten times higher than C-PQDR and more than 13 times higher than G-PQDR.

VI. DISCUSSION

We can notice that the compression rate depends mainly on two characteristics, the signal duration and the number of disturbances present in the signal. The longer the duration of the analyzed signal, the higher the CR value. Long-duration synthetic signal tests showed that CR could reach the order of thousands to one when dealing with long signals. On the other hand, the more disturbances present in the signal, the more novelties are detected and, consequently, the lower the CR value.

In addition, during the sampling rate variation tests, signals containing at least one disturbance every 1.16 s were used, which is a considerably high occurrence rate. Even so, the proposed method showed a good compression rate and excellent quality of reconstruction.

The SVDSCS compression is performed in real-time. Any sampling frequency is allowed, limited by the hardware specifications. On the other hand, the reconstruction process is performed offline. Thus, it does not influence the complexity of the hardware implementation.

Finally, it is important to compare the advantages of the proposed SVDSCS with some of its limitations. The main advantages are higher CR values and better reconstruction fidelity. On the other hand, the reconstruction of very abrupt

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Eder B. Kapisch received the B.S., M.Sc., and Ph.D. degrees in electrical engineering from the Federal University of Juiz de Fora (UFJF), Juiz de Fora, Brazil, in 2013, 2015, and 2019, respectively.

In 2017 and 2018, he was a Ph.D. Researcher with the TU/e (Eindhoven University of Technology), Eindhoven, The Netherlands. He is currently a Professor with the Department of Electrical Engineering, UFJF. Some of his works involve the development and implementation of digital signal processing algorithms using FPGA platform applied to detection and compression of power quality electrical disturbances. His research interests include electronic systems, electronic instrumentation, and digital signal processing applied to power systems.



Victor V. de Moraes received the B.S. degree in electrical engineering from the Federal University of Juiz de Fora (UFJF), Juiz de Fora, Brazil, in 2018. He is currently working toward the M.Sc. degree in electrical engineering—biomedical engineering with the State University of Campinas, Campinas, Brazil.

His current research interests include implementation of digital signal processing algorithms in FPGA platform applied to power quality compression and biomedical signal processing.



Leandro R. M. Silva (Member, IEEE) received the B.S., M.Sc., and the Ph.D. degrees in electrical engineering from the Federal University of Juiz de Fora (UFJF), Juiz de Fora, Brazil, in 2009, 2012, and 2016, respectively.

He is currently a Professor with the Department of Electrical Engineering, UFJF. He also works with computational intelligence, load modeling, phasor measurement units, time-varying harmonic estimation, power quality signals compression, and embedded systems. His research interests include real-time digital signal processing algorithms development and implementation on dedicated platforms, such as digital signal processor and FPGA, applied to power systems.



Luciano M. A. Filho received the B.S. degree in electrical engineering from the State University of Rio de Janeiro, Rio de Janeiro, Brazil, in 2001, the M.Sc. degree in scientific instrumentation from the Brazilian Center for Research in Physics, Rio de Janeiro, in 2004, and the Ph.D. degree in electrical engineering from the Federal University of Janeiro/COPPE, Rio de Janeiro, in 2009.

He is currently a Professor with the Department of Electrical Engineering, Federal University of Juiz de Fora, Juiz de Fora, Brazil. Part of his Ph.D. research was conducted at the European Council for Nuclear Research, Geneva, Switzerland. His research interests include signal processing for calorimetry, digital processing in FPGA, and power quality.



Carlos A. Duque (Senior Member, IEEE) received the B.S. degree from the Federal University of Juiz de Fora (UFJF), Juiz de Fora, Brazil, in 1986, and the M.Sc. and Ph.D. degrees from the Catholic University of Rio de Janeiro, Rio de Janeiro, Brazil, in 1989 and 1997, respectively, all in electrical engineering.

He is currently a Full Professor with the Department of Electrical Engineering, UFJF. He was a Visitor Researcher with the Center for Advanced Power Systems, Florida State University, Tallahassee, FL, USA, where he conducted his Postdoctoral research, in 2007 and 2008. His research interests include power quality analysis, digital instrumentation, and digital signal processing.